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Probing Local Thermal Conductivity Variations in CVD Diamond with Large Grains by Time-Domain Thermoreflectance

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ABSTRACT

Chemical vapor deposited (CVD) diamond, due to its high thermal conductivity, is an attractive candidate for thermal management of GaN-based high-electron mobility transistors (HEMTs). However, because of its heterogeneous grain structure, CVD diamond has a spatially inhomogeneous thermal conductivity at the microscale. To understand this inhomogeneity and the effect of structural imperfections on thermal conduction, time-domain thermoreflectance (TDTR) is used to study the local thermal conductivity of two samples: a heavily boron-doped ~534 μm-thick diamond sample with an average surface grain size of ~23 μm, and an undoped diamond sample that was cut from a bulk piece of CVD diamond. For the doped diamond, large thermal conductivity variations (of nearly 50 %) are observed across the surface of the sample. For the undoped sample, the large average grain size (several hundred μm) results in a high local thermal conductivity (>2000 W/m-K, close to the conductivity of bulk diamond). The thermal conductivity is not seen to change significantly with grain size (127 - 260 μm), and we measure up to ~8 % variation in the local thermal conductivity. We speculate that grain boundary scattering affects phonon transport differently in the two samples, possibly due to varying amounts of near-boundary disorder. This work provides insights to understand the local thermal conductivity inhomogeneity and phonon transport across grain boundaries in CVD diamond with large grains, which is important for thermal management applications in high-power electronics.

KEY WORDS: Electronics cooling, Conduction, Thermal management and control, CVD diamond, Time-domain thermoreflectance, Inhomogeneous thermal conduction.

1. INTRODUCTION

With the development of GaN-based high-electron mobility transistors (HEMTs), the localized power densities in GaN HEMTs can exceed 90 kW/cm^2, resulting in high channel temperatures that degrade device performance and reliability [1, 2]. The integration of high thermal conductivity CVD diamond into GaN HEMTs allows for effective heat spreading and the ability to keep these devices cool [3-9]. Because of its heterogeneous grain structure, polycrystalline CVD diamond shows inhomogeneous thermal conductivity [10-12]. The inhomogeneous heat conduction in CVD diamond has attracted significant attention because it can have an important impact on heat dissipation through thermal conduction [10, 13-20]. However, most

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previous work has focused on measuring the "effective" thermal conductivity of thin films, and did not consider the effect of grain structure on local thermal conductivity variations for CVD diamond with large grains. Recently, Sood *et al.* reported the first direct measurements of spatially-varying thermal conductivity in large-grained CVD diamond using the scanning time-domain themroreflectance (TDTR) technique [21]. TDTR is an optical pump-probe thermal characterization tool that is used to measure thermal properties of both bulk and nanostructured materials. A modulated pump beam heats the sample surface periodically and a delayed probe beam measures the temperature decay of the sample surface through a change in thermoreflectance. By fitting the experimental signal picked up by a lock-in amplifier with a multi-layer thermal model, thermal properties of the sample can be extracted [7]. The TDTR beam spot sizes are several or tens of micrometers depending on the magnification of the objective used. The grain sizes of thick CVD diamond samples are typically much larger than both the TDTR spot sizes and thermal penetration depths. The thermal conductivity probed by TDTR is a local thermal conductivity that is sensitive to the diamond sampled by the laser spot. For that part of the diamond, we consider its local thermal conductivity as an isotropic effective thermal conductivity. To understand the inhomogeneous thermal conduction in the thick diamond samples, TDTR can be used to map the thermal conductivity distribution within and across these large grains by integrating a motion-controlled sample stage into the TDTR system.

In this work, we map the thermal conductivity variation of the same sample in Ref. [21] to study the inhomogeneous thermal conduction. This sample (Sample 1) is a ~534 μm-thick CVD diamond that is heavily boron-doped (~10^{21} cm^{-3}) with an average grain size of ~23 μm. Following an approach similar to Ref. [21], we obtain a 2D map of its inhomogeneous thermal conductivity. Furthermore, to study the relation of local thermal conductivity and grain size in diamond samples with large grains (grain size over 100 μm), we use traditional TDTR to measure the local thermal conductivity of an undoped sample (Sample 2) cut from a bulk CVD diamond sample with variable grain sizes over a distance of 1 cm. We also measure the local thermal conductivity by full TDTR scans with a step size smaller than the grain size over a distance of 300 μm to study the local thermal conduction across grain boundaries in high-quality diamond.

2. SAMPLE DESCRIPTION AND CHARACTERIZATION

The schematic diagram of our TDTR setup is shown in Fig. 1. A modulated pump beam heats the sample surface periodically while a delayed probe beam detects the temperature of the sample surface via thermoreflectance. We deposited a ~90 nm thick layer of Al on the sample surface to act as a transducer. The diamond sample was mounted on a positioning stage (Brushed Motor Controller KDC 101) which moves along X and Y directions as shown in Fig. 1. The pump and probe beam radii in this work are measured by a DataRay scanning slit beam profiler.

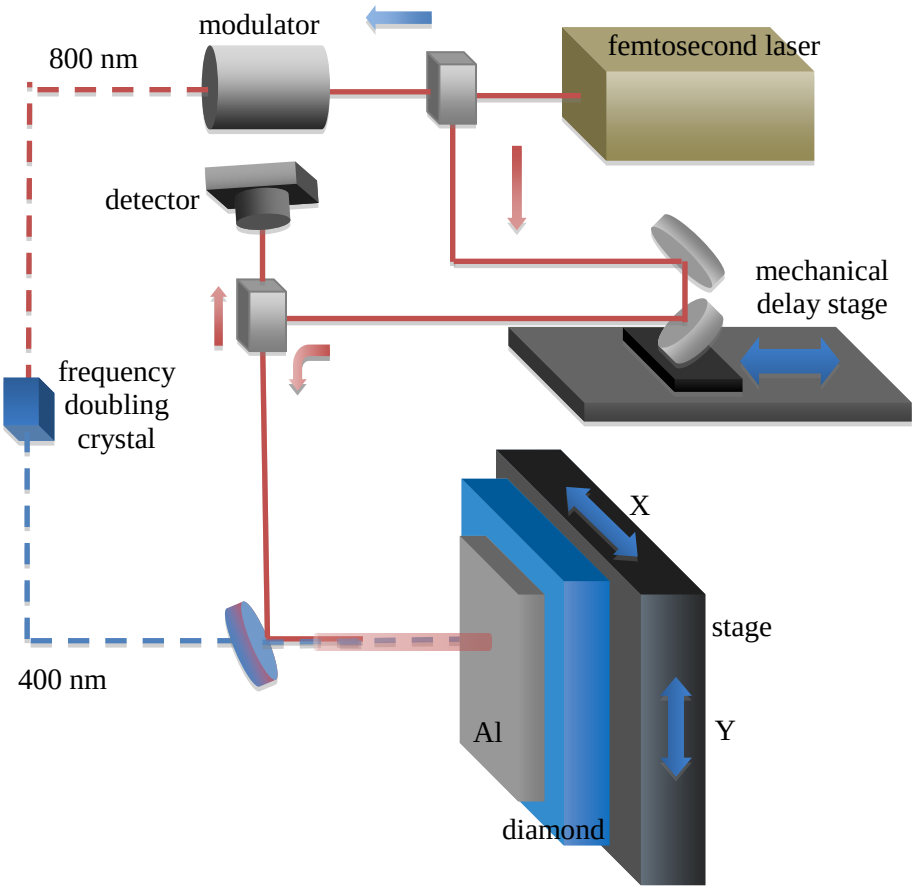


Fig. 1. Schematic diagram of TDTR system and the sample stage. Pump (blue) beam heats the sample and probe (red) beam measures the temperature variation.

By moving the sample stage automatically, we map the region on the diamond sample which has been characterized by EBSD. Instead of performing a TDTR measurement at every spot on the sample, we record the TDTR ratio after fixing the delay time at a point where the sensitivity of Al-diamond thermal interface conductance is zero. [21] The definition of TDTR sensitivity is given by Equation (1)

$$S_i = \frac{\partial \ln(-V_{in}/V_{out})}{\partial \ln(p_i)} \quad (1)$$

where S_i is the sensitivity to parameter p_i, and $-V_{in}/V_{out}$ is the TDTR ratio signal. If the sensitivity to a certain parameter is near-zero, the variation of that parameter does not affect the TDTR signal. We chose our probe delay time to minimize the sensitivity to the Al-diamond thermal interface conductance, ensuring that measured variations in the TDTR ratio are primarily due to variations in the diamond thermal conductivity. After performing a full TDTR measurement, we obtain both the TDTR ratio at that delay time and the local thermal conductivity after fitting the data. To help align the thermal conductivity map with the grain orientation map, four markers were placed at the corners of the mapping region as reference features. Cuts were made into the surface of the diamond using focused ion beam (FIB) milling.

The schematic diagram of the undoped sample (Sample 2) is shown in Fig. 2. The sample is a 290-μm-thick diamond polished with an angle of 5.9 degrees from a diamond slab. The top side of the slab has larger grains than the bottom side. By polishing the slab with an angle, we obtain the undoped sample with a variable grain size along the x axis. The average grain sizes with x=2 to x=12 mm range from ~260 μm to ~127 μm, which is much larger than that of Sample 1 (~23 μm).[22] This sample was not doped.

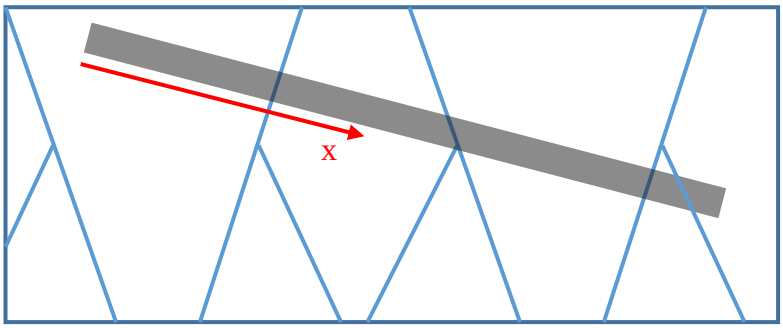


Fig. 2. The schematic diagram of the undoped sample (Sample 2).

3. RESULTS AND DISCUSSION

Fig. 3(a) shows the color-scale in-plane grain-orientation map measured by EBSD. Blue and red denote (111) and (100) orientation, while green denotes (110) orientation. The data is taken from Ref. [21]. Fig. 3(b) shows the TDTR thermal conductivity map of the doped sample. The pump and probe spot sizes (diameters) in this scan are 8.3 µm and 6.8 µm, respectively, and step-size is 10 µm. The thermal conductivity map has a lower resolution than that in Ref. [21] because of the larger step distance and spot-size. As shown in Fig. 3(b), we observe large thermal conductivity variations across the mapped area. We find that at the center of large grains, the peak thermal conductivity is ~1000 W/m-K, lower than that of a typical pure single-crystal. The high doping level in this sample contributes to lowering of the overall thermal conductivity. Within small grains and near grain boundaries, we observe a significant reduction in local thermal conductivity. We hypothesize that this is likely a result of additional structural disorder in the immediate vicinity of grain boundaries, resulting in strong phonon scattering and reduced mean free paths.

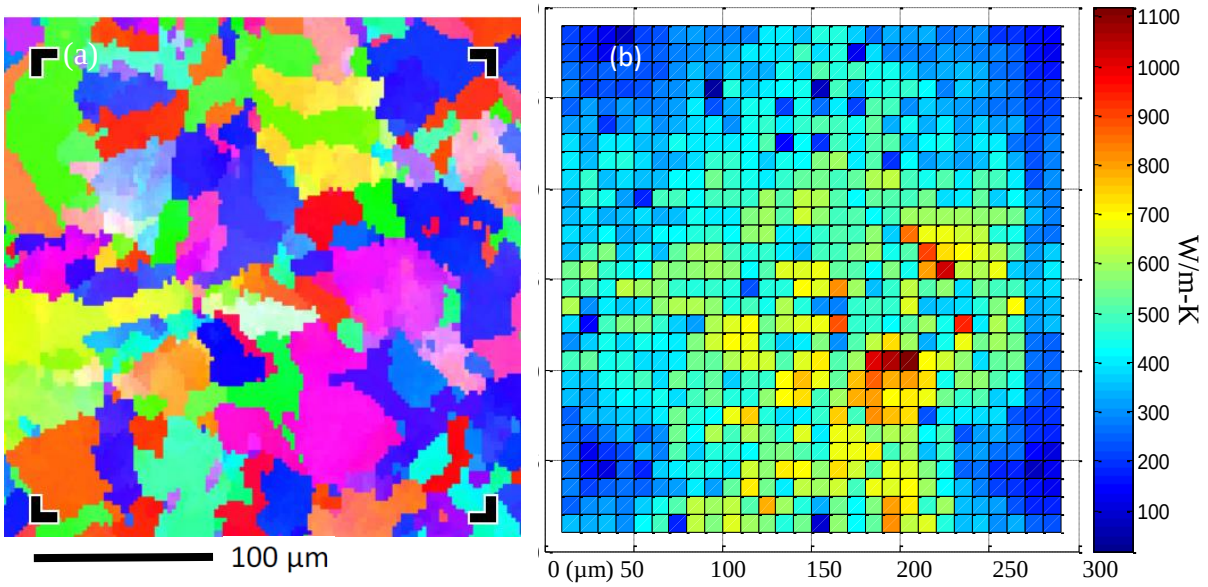


Fig. 3. (a) EBSD in-plane grain orientation map. (data taken from Ref. [21]) (b) Thermal conductivity map of Sample 1.

Fig. 4 shows the local thermal conductivity measured by TDTR along the x axis on Sample 2 (i.e. the undoped sample). The 3-omega results measured on the same sample in Ref. [22] are also included for comparison. In general, the TDTR and 3-omega results are consistent. The measured thermal conductivities are very high (>2000 W/m-K), which are close to that of single crystal diamond. The measured thermal

conductivity does not change with grain size because the grain size is much larger than phonon mean free path in diamond. For very large grain sizes, lattice quality, instead of grain boundary scattering, predominantly affects thermal conductivity. This high-quality diamond facilitates heat spreading for electronics. As shown in Fig. 4, the TDTR thermal conductivity values have larger variation than those values measured by 3-omega because TDTR probes more localized thermal properties than 3-omega. The spot sizes of TDTR used in these measurements are only 22.3 µm and 11.8 µm (pump and probe radii). On the other hand, the length of the 3-omega heater line is 1235 µm and the probed depth is ~150 µm.[22] The heater length is much larger than the average grain size, which means that the heater line crosses several grains, and therefore, the measured thermal conductivity is an average value. For spots measured by TDTR at x=1 and 11 mm, we believe the thermal conductivity is slightly higher as these are likely located at the centers of large grains, while for the spot at x=9 mm, it is likely located within a small grain or close to a grain boundary. For applications in electronics thermal management, determining local thermal conductivity is key to affecting device performance, and TDTR is a valuable technique to probe these spatial variations.

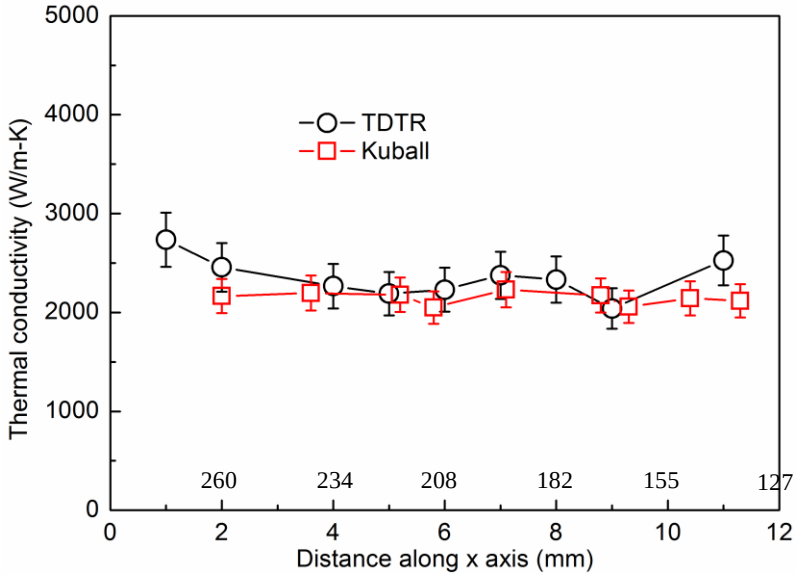


Fig. 4. The local thermal conductivity along x axis measured by TDTR. The 3-omega (Kuball) results in Ref. [22] are also included for comparison. The number above the horizontal axis is the average grain size in units of µm.[22]

To probe the local thermal conductivity variation within a grain and across grain boundaries in this high-quality diamond (Sample 2), we measured the local thermal conductivity along a line parallel to the x axis. We moved the laser spot close to x=8 µm and measured the local thermal conductivity of diamond by full TDTR scans. The distance between two spots is 50 µm. The total scan length is 300 µm while the average grain size of this part of the diamond is ~175 µm. Therefore, our measured area likely covers one or more grain boundaries. The local thermal conductivity is plotted in Fig. 5(a). The average thermal conductivity of these spots is almost the same as the 3-omega results. But TDTR captures the variation of local thermal conductivity. The local thermal conductivity at x=8.3, 8.35, and 8.5 mm is about 15% larger than at x=8.2, 8.25, and 8.4 mm. We speculate that the spots at 8.3, 8.35, and 8.5 mm are in the large grains while the spots at x=8.2, 8.25, and 8.4 mm are near grain boundaries. As shown in Fig. 5(b), we plot the normalized local thermal conductivity of the doped sample and the undoped sample (normalized by their average value along lines). The line for the doped sample is vertical line in Fig. 3 (at x =160 µm). This line covers more than ten grains while the line for the undoped sample only covers one or two grain boundaries. The thermal conductivity variation of the doped sample is much larger than that of the undoped sample (only ~8%). One possible reason is that the large spot size averages the thermal conductivity of a large part of the high-quality diamond near the grain boundary. Another possible reason is the differences in the phonon transmittance between the two samples. This could be a result of the higher degree of doping in Sample 1, which likely causes additional structural disorder in the immediate vicinity of the grain boundary. Understanding the role

of point defects and near-interface disorder on phonon transport across grain boundaries is a topic that deserves additional study, and will be the subject of future work.

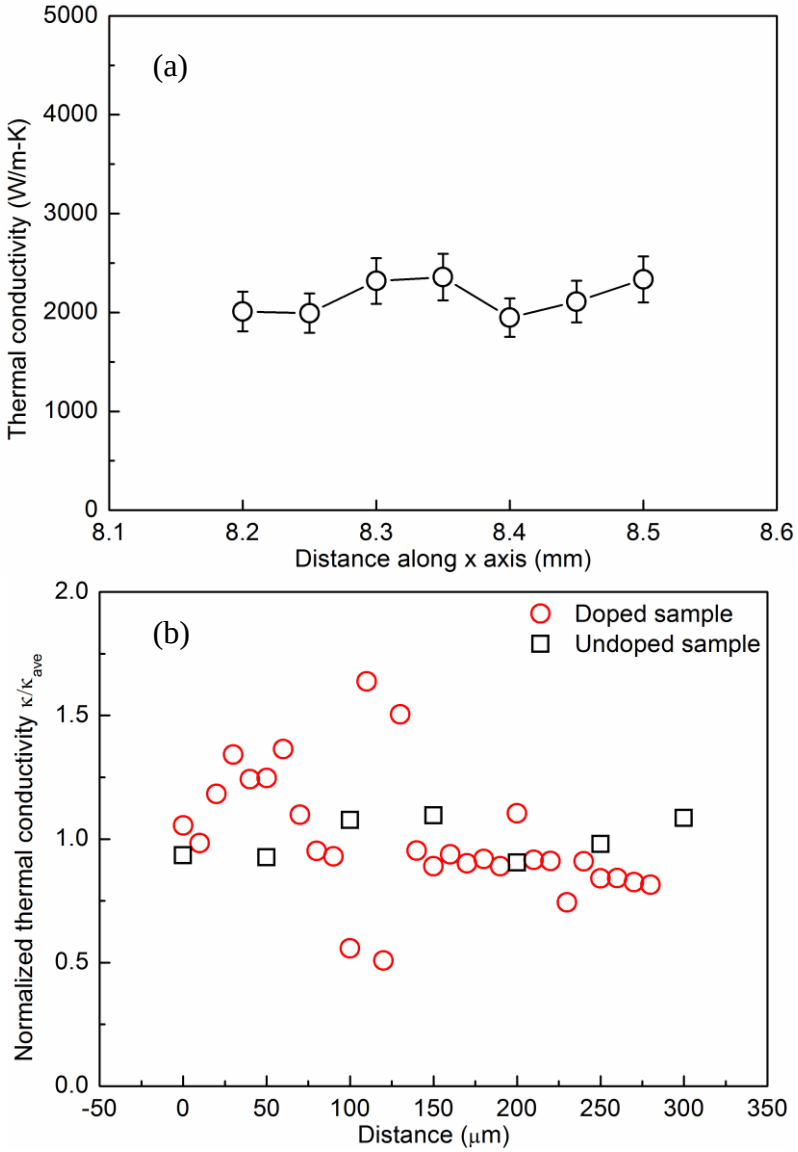


Fig. 5. (a) The local thermal conductivity along x axis measured by full TDTR scans with fine step size (50 μm) in Sample 2. The measured area is different from that of Fig. 4. The average grain size near the measured area is 175 μm. (b) The normalized local thermal conductivity (normalized by their average values) are plotted to compare the local thermal conductivity variations between the doped sample and the undoped sample.

4. CONCLUSION

CVD diamond, due to its high thermal conductivity, is an attractive candidate to dissipate the localized self-heating in GaN-based HEMTs which has an important effect on device performance and reliability. However, because of its inhomogeneous microstructure, polycrystalline diamond has an inhomogeneous thermal conductivity at the microscale. In our work, we map the thermal conductivity of a heavily doped diamond with large grains. We observed large thermal conductivity variations across the sample. High thermal conductivity was observed at the center of large grains and low thermal conductivity was observed at small grains and near grain boundaries due to increased phonon-grain boundary scattering. Additionally, we used TDTR to measure the local thermal conductivity of an undoped sample with very large grains. The large average grain size (several hundred μm) results in large local thermal conductivity (>2000 W/m-K,

close to thermal conductivity of bulk diamond) and the conductivity does not change with grain size (127 - 260 μm). TDTR shows good capability to probe local thermal conductivity variations and small thermal conductivity variation (~8%) was observed in the high-quality undoped sample. Our results point towards the important role of near-boundary disorder in affecting phonon transmission across grain boundaries, and consequently, their impact on thermal transport. This work provides insights to understand the local thermal conductivity inhomogeneity and phonon transport across grain boundaries in CVD diamond with large grains, which is important for thermal management of high power electronics.

ACKNOWLEDGMENT

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NOMENCLATURE

α_t	thermal diffusivity	(m^2/s)
f	modulation frequency	(MHz)
S_i	TDTR sensitivity to parameter i	
$-V_{in}/V_{out}$	TDTR ratio	
p_i	the value of parameter i	

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